

Compound assignment / short hand operators:

Here we are using assignment operator with the combination of other operators as follows.

+=, -=, *=, %=, /=, ^=, <<=, >>=,...

Eg:

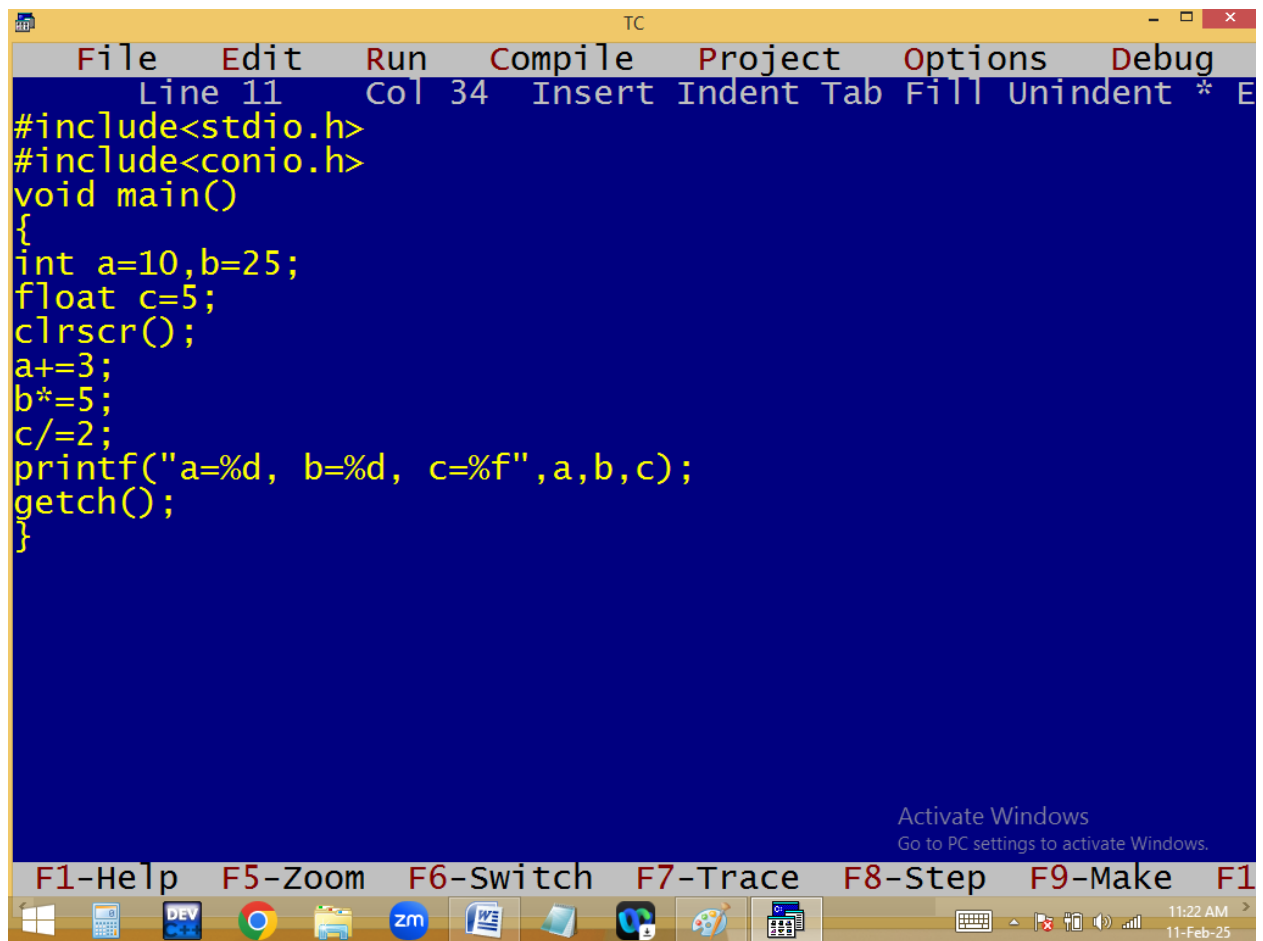
int a=10, b=25;

float c=5;

a+=3; i.e. a=a+3 → 10+3=13

b*=5; i.e. b=b*5 → 25*5 =125

c/=2; i.e. c=c/2 → 5/2 = 2.500000

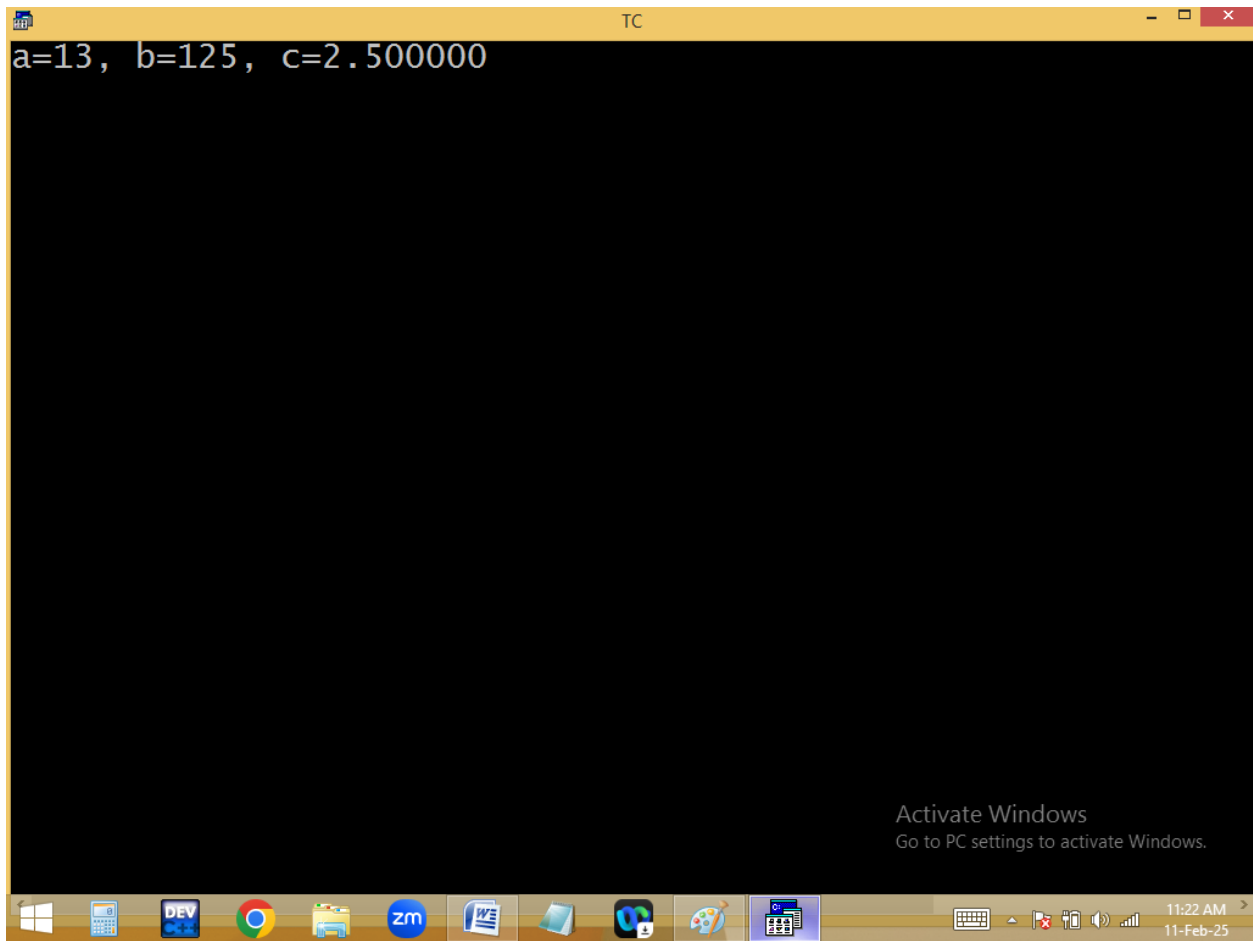


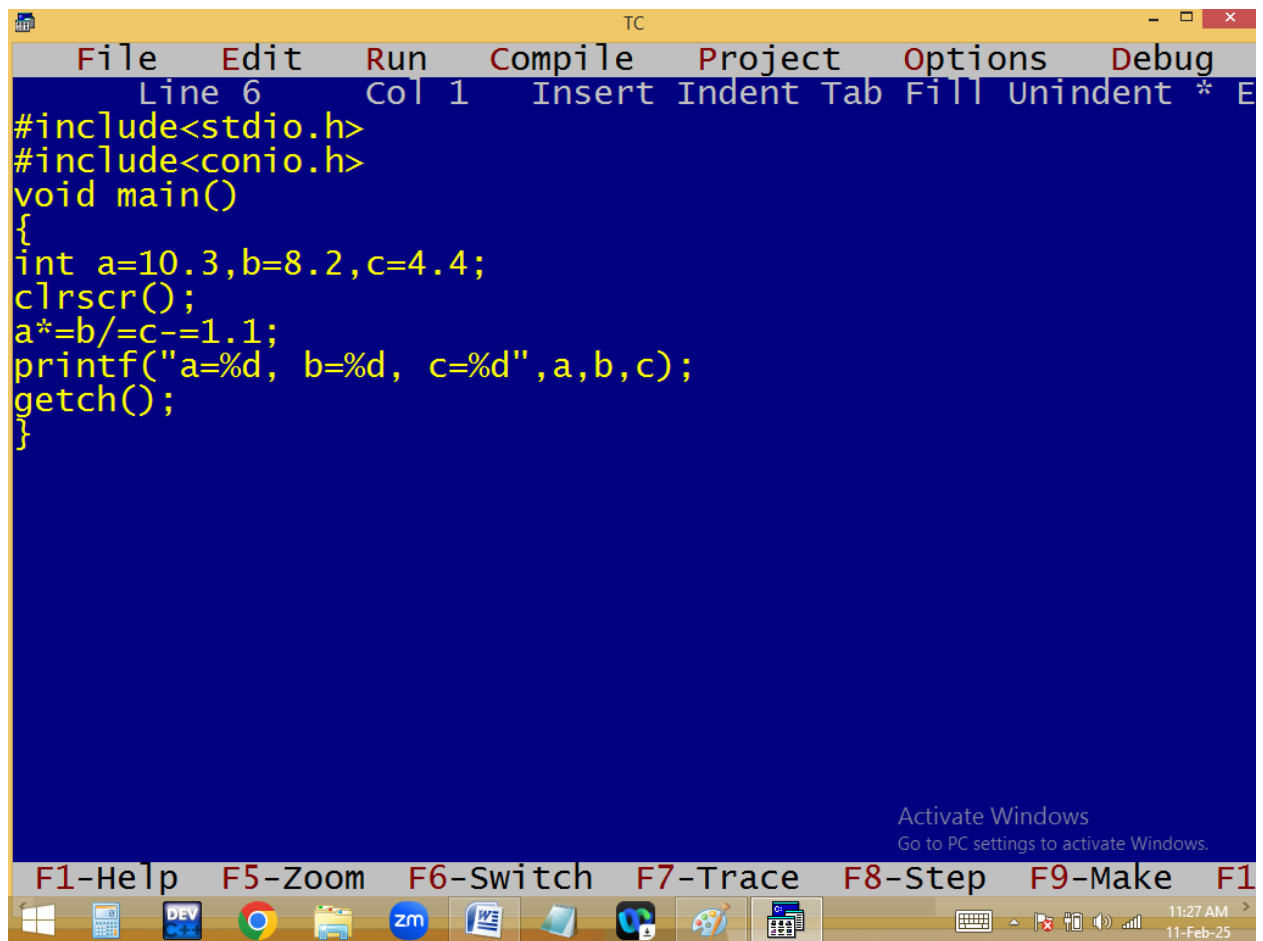
```
TC
File Edit Run Compile Project Options Debug
Line 11 Col 34 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10,b=25;
float c=5;
clrscr();
a+=3;
b*=5;
c/=2;
printf("a=%d, b=%d, c=%f",a,b,c);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

11:22 AM
11-Feb-25



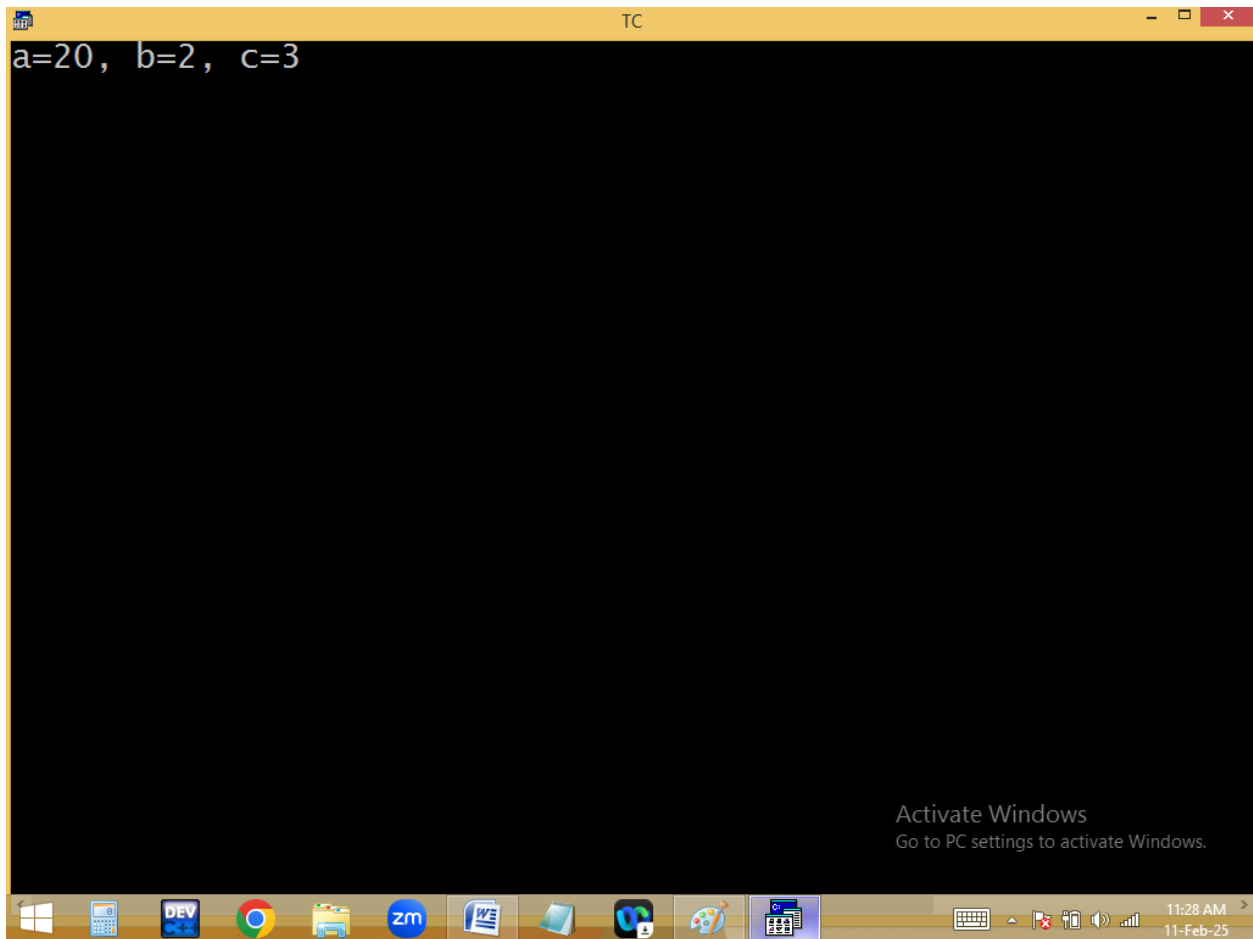


```
TC
File Edit Run Compile Project Options Debug
Line 6 Col 1 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10.3,b=8.2,c=4.4;
clrscr();
a*=b/=c-=1.1;
printf("a=%d, b=%d, c=%d",a,b,c);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

11:27 AM
11-Feb-25



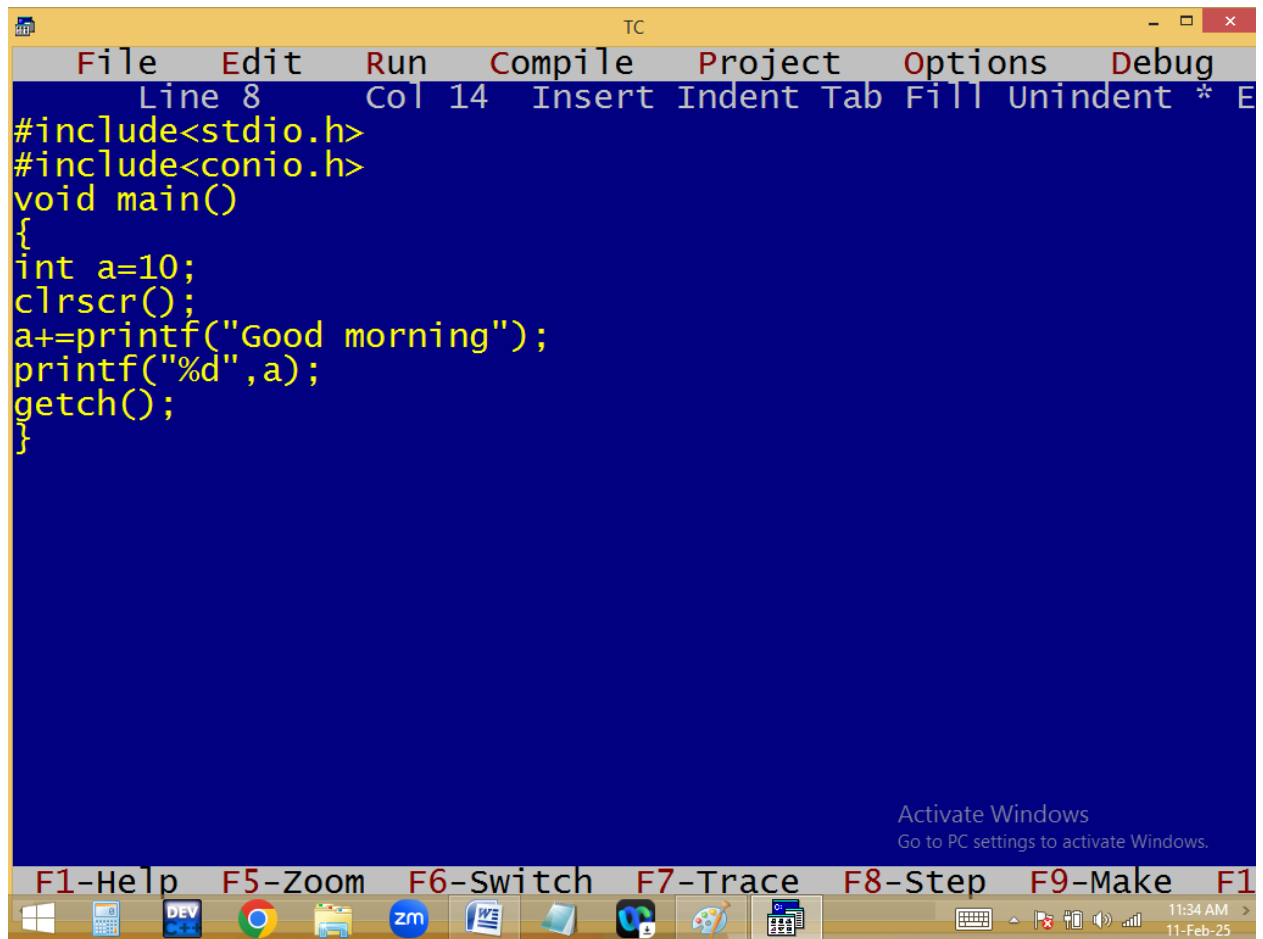
```
int a=10, b=8, c=4;
```

`a*=b/=c=1.1;`

$c=1.1 \implies c=c-1.1 \implies 4-1.1 = 2.9$
↓
3 ✓

$b/=c \implies \text{i.e. } b=b/c \implies 8/3 \implies 2 \checkmark$

$a*=b \implies \text{i.e. } a=a*b \implies 10 * 2 = 20 \checkmark$

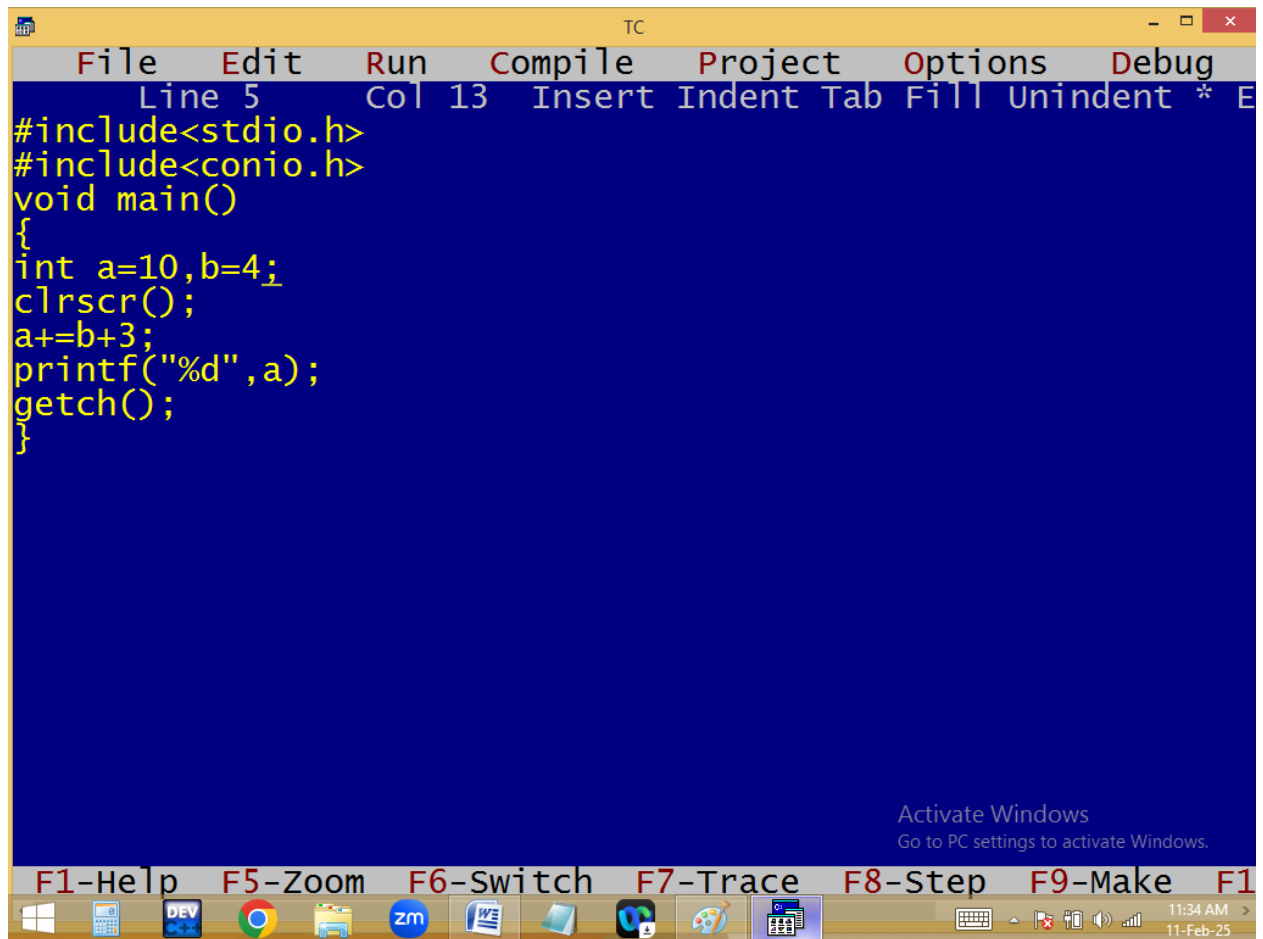


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 8 Col 14 Insert Indent Tab Fill Unindent * E". The code editor contains the following C program:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10;
clrscr();
a+=printf("Good morning");
printf("%d",a);
getch();
}
```

At the bottom of the IDE window, there is a text box that says "Activate Windows Go to PC settings to activate Windows." Below the IDE window is the Windows taskbar. It features a row of function key shortcuts: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F10-Exit. The taskbar also includes icons for the Start menu, Task View, File Explorer, Google Chrome, Zoom, Word, and other applications. The system tray on the right shows the time as 11:34 AM and the date as 11-Feb-25.

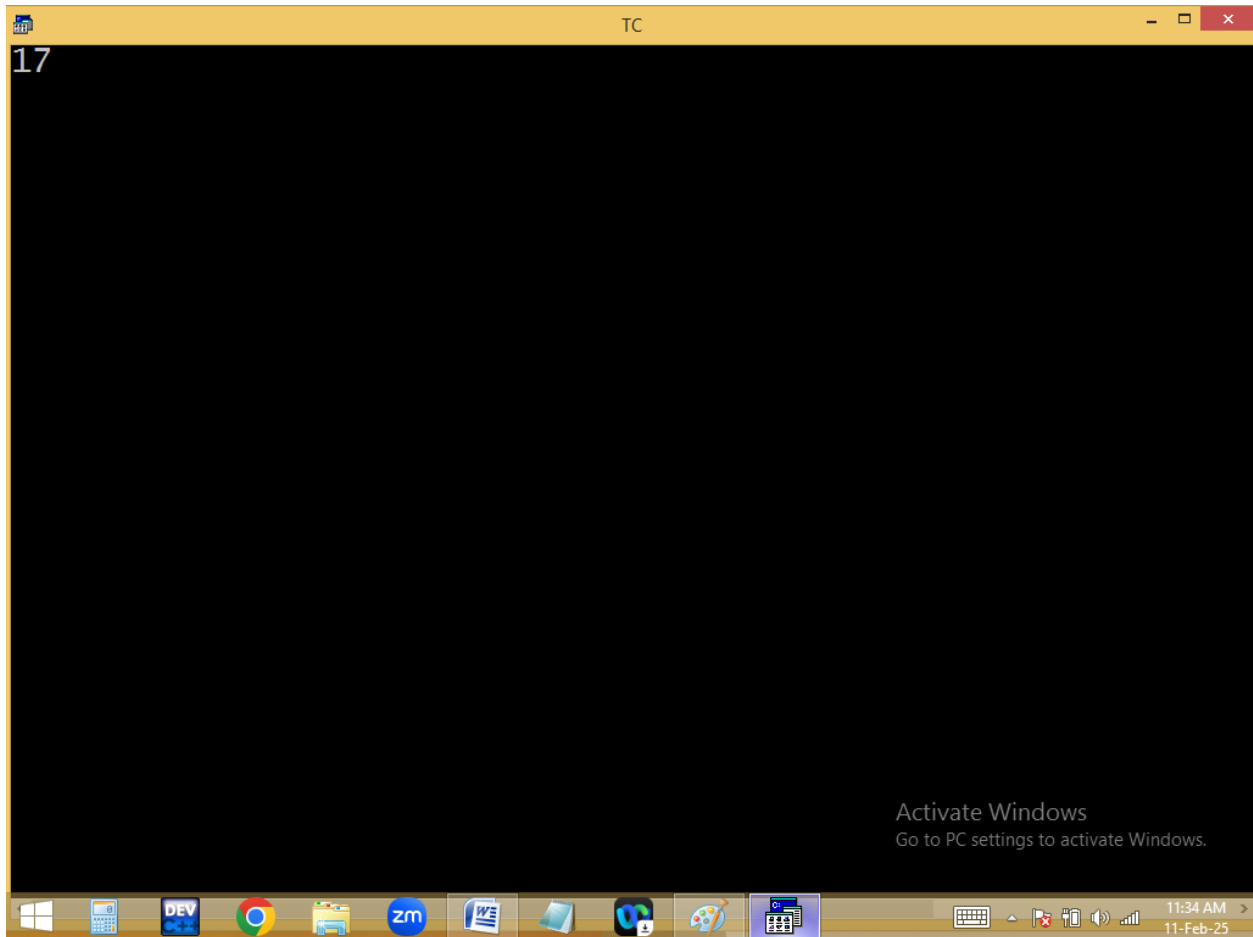




The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 5" and "Col 13". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10,b=4;
clrscr();
a+=b+3;
printf("%d",a);
getch();
}
```

At the bottom of the IDE window, there is a text prompt: "Activate Windows Go to PC settings to activate Windows." Below the IDE window is the Windows taskbar, which includes icons for the Start menu, Task View, File Explorer, Microsoft Edge, Zoom, Word, and several other applications. The system tray on the right shows the time as 11:34 AM and the date as 11-Feb-25. A row of function key shortcuts is visible just above the taskbar: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F10-Exit.



a=10, b=4

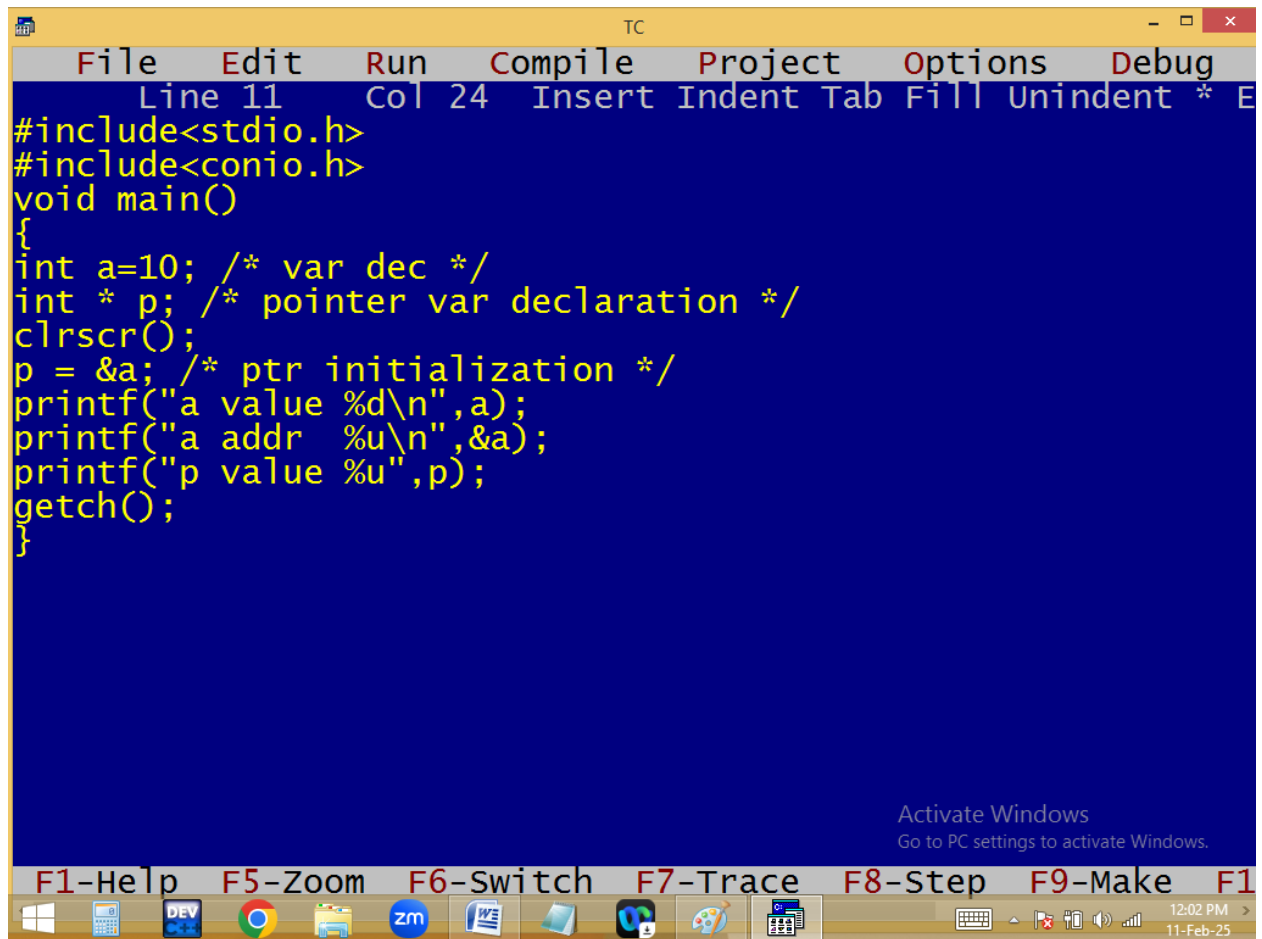
a+= b+3 ==> b+3 ==> 4+3 ==> 7

a=a+7 ==> 10+7 = 17

Address operators:

1. & - address of the variable

2. * - pointer – stores the address of another variable

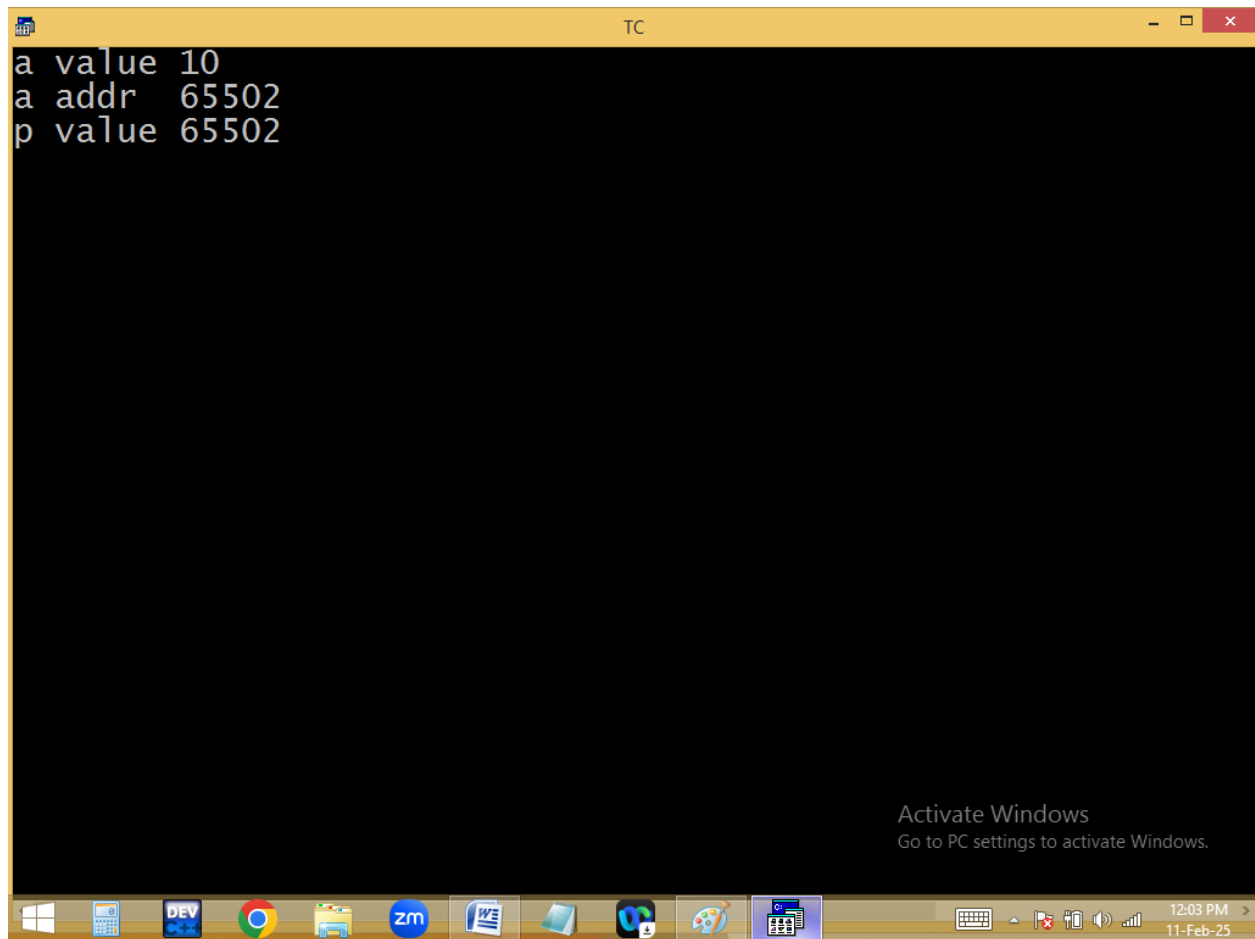


```
TC
File Edit Run Compile Project Options Debug
Line 11 Col 24 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10; /* var dec */
int * p; /* pointer var declaration */
clrscr();
p = &a; /* ptr initialization */
printf("a value %d\n",a);
printf("a addr %u\n",&a);
printf("p value %u",p);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

12:02 PM
11-Feb-25

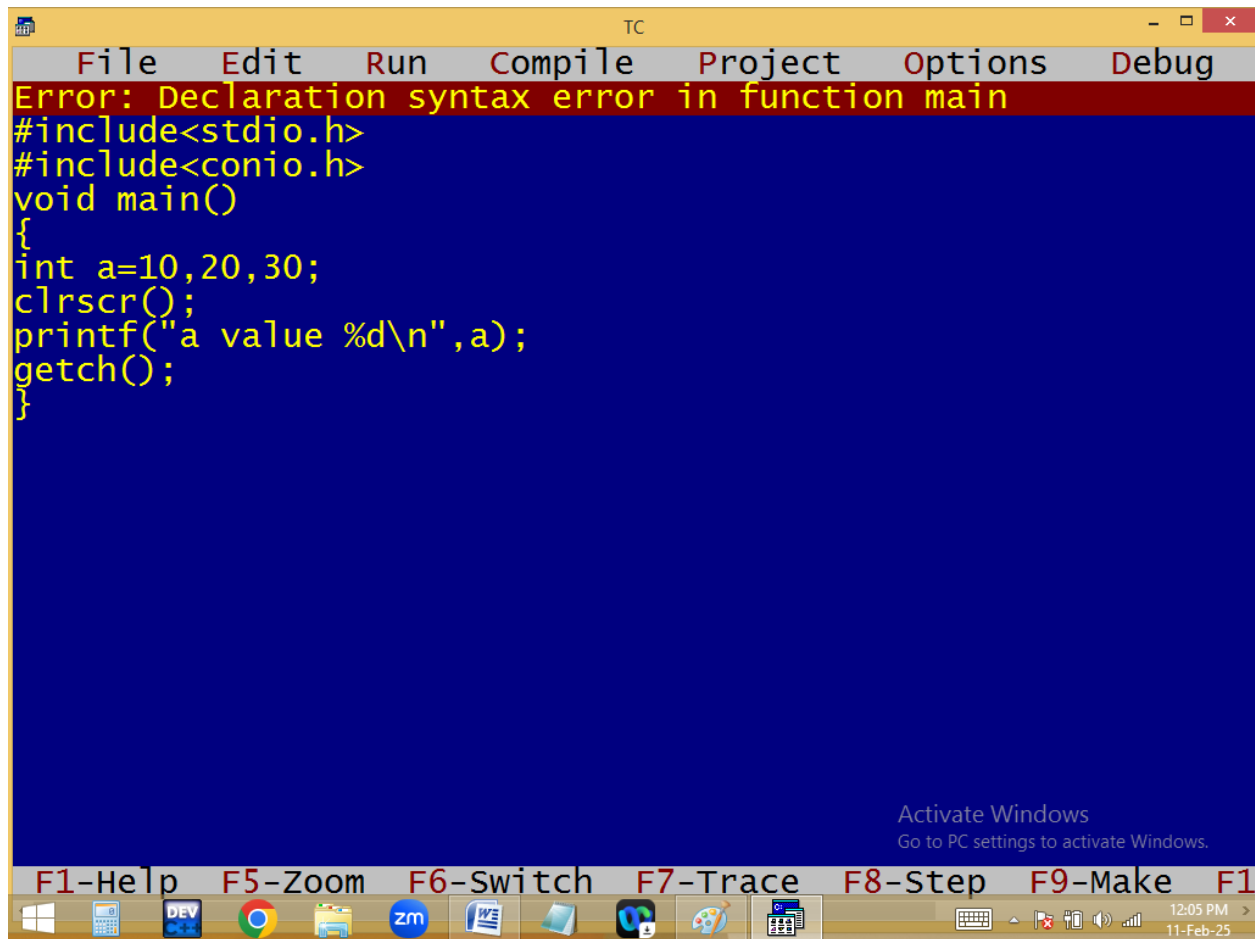


The image shows a Windows 10 desktop environment. A terminal window titled "TC" is open, displaying the following assembly code:

```
a value 10  
a addr 65502  
p value 65502
```

The desktop background is black. In the bottom right corner, there is a watermark that says "Activate Windows" and "Go to PC settings to activate Windows." The taskbar at the bottom contains several icons: Windows Start button, Task View, File Explorer, Microsoft Edge, Zoom, Word, PowerPoint, OneDrive, and a folder icon. The system tray on the right shows the time as 12:03 PM and the date as 11-Feb-25.

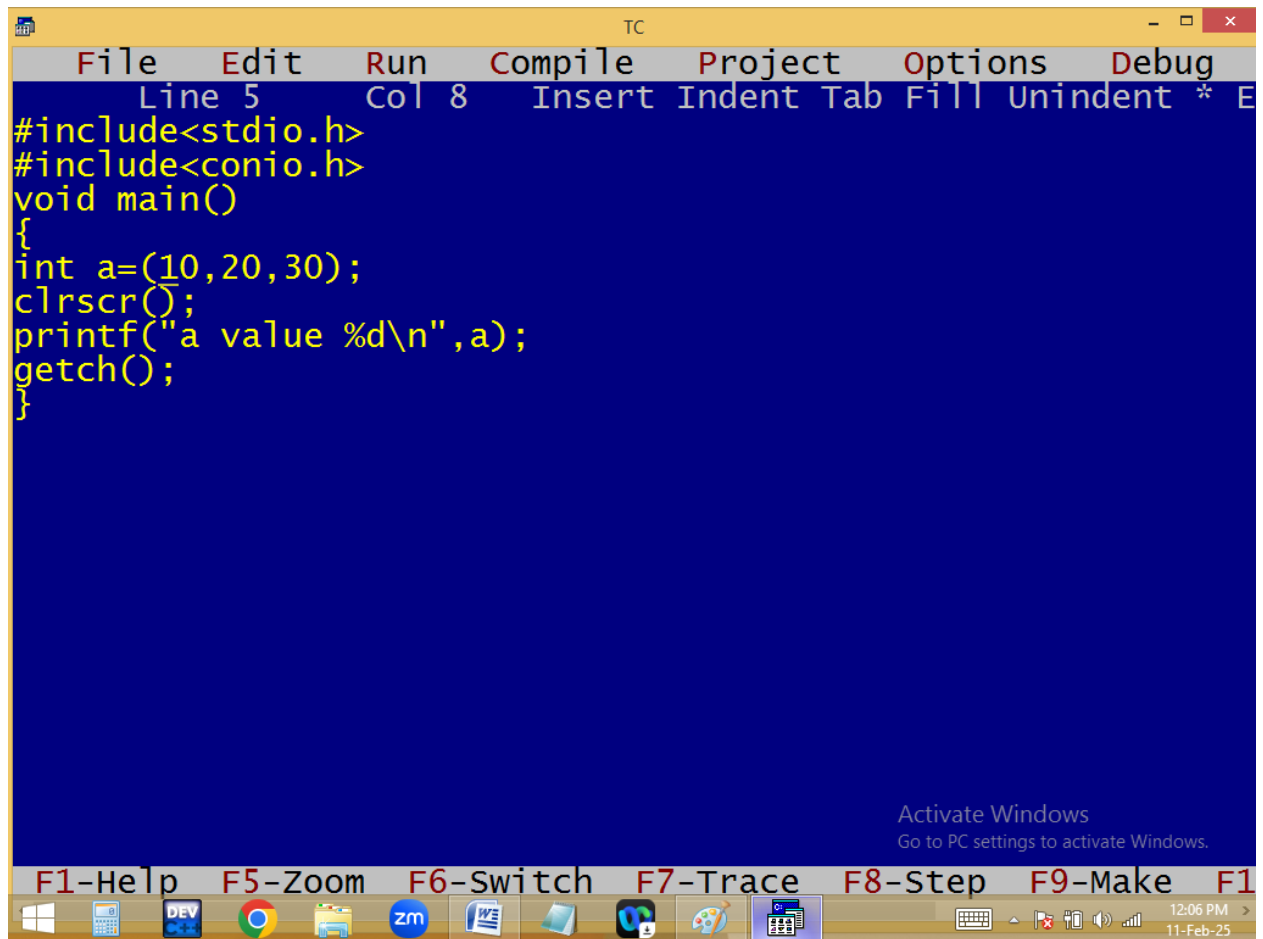
() and , separators:



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error message banner at the top reads "Error: Declaration syntax error in function main". The code editor contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10,20,30;
clrscr();
printf("a value %d\n",a);
getch();
}
```

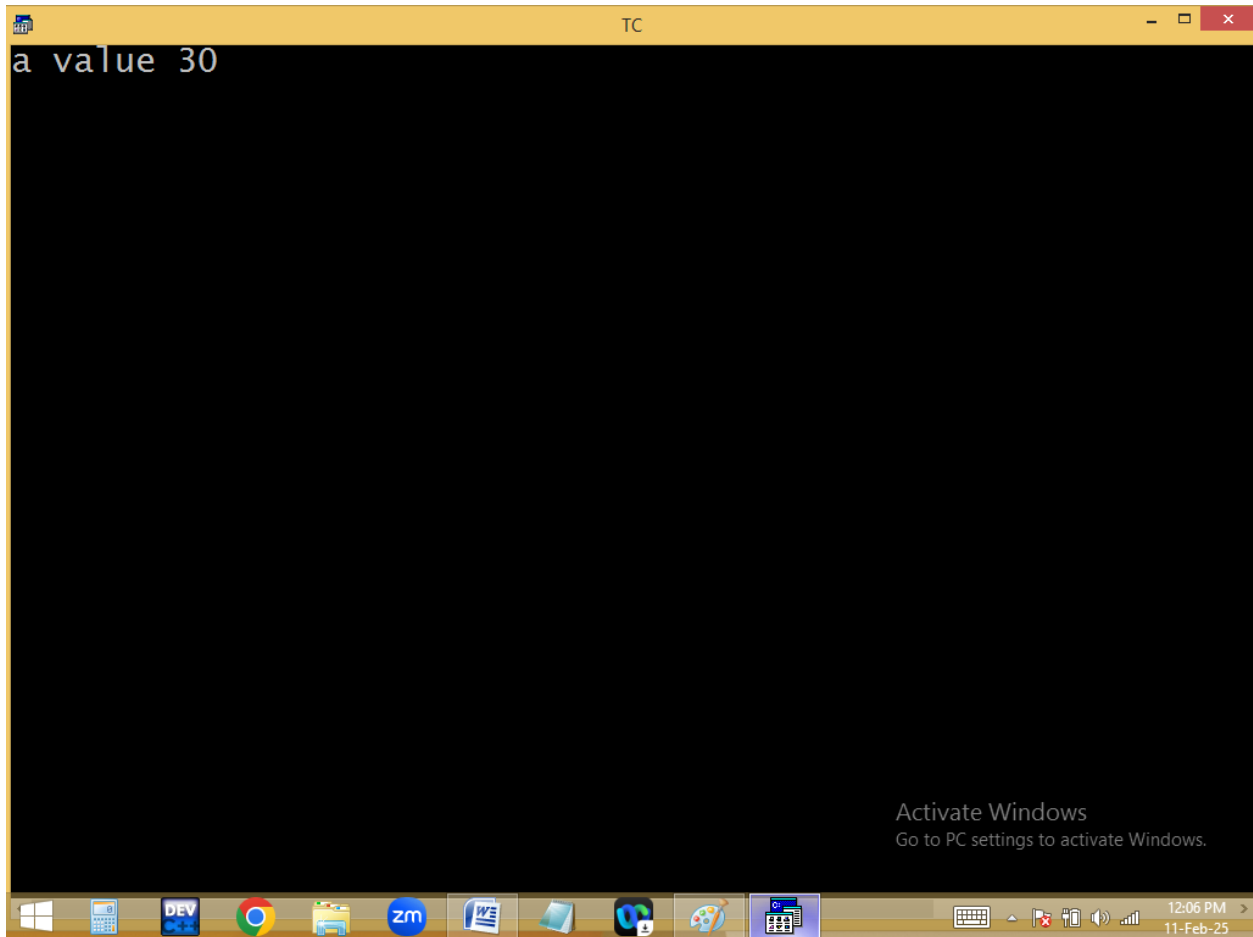
At the bottom of the IDE window, there is a text prompt: "Activate Windows Go to PC settings to activate Windows." Below the IDE window is the Windows taskbar. It features a row of function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Exit". The taskbar also includes icons for the Start menu, Task View, File Explorer, Google Chrome, and several other applications. The system tray on the right shows the time as "12:05 PM" and the date as "11-Feb-25".



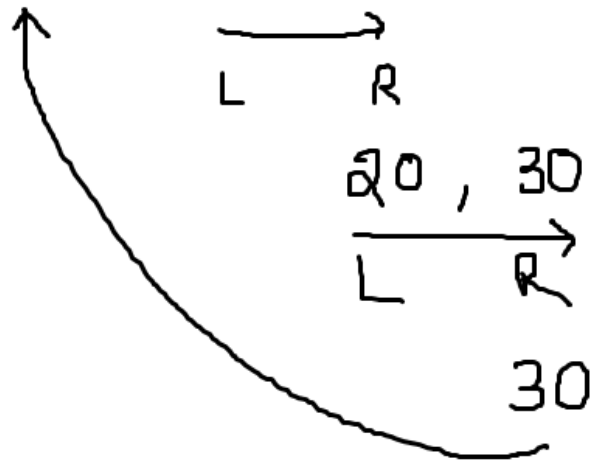
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 5", "Col 8", and lists editing actions: Insert, Indent, Tab, Fill, Unindent, and *. The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=(10,20,30);
clrscr();
printf("a value %d\n",a);
getch();
}
```

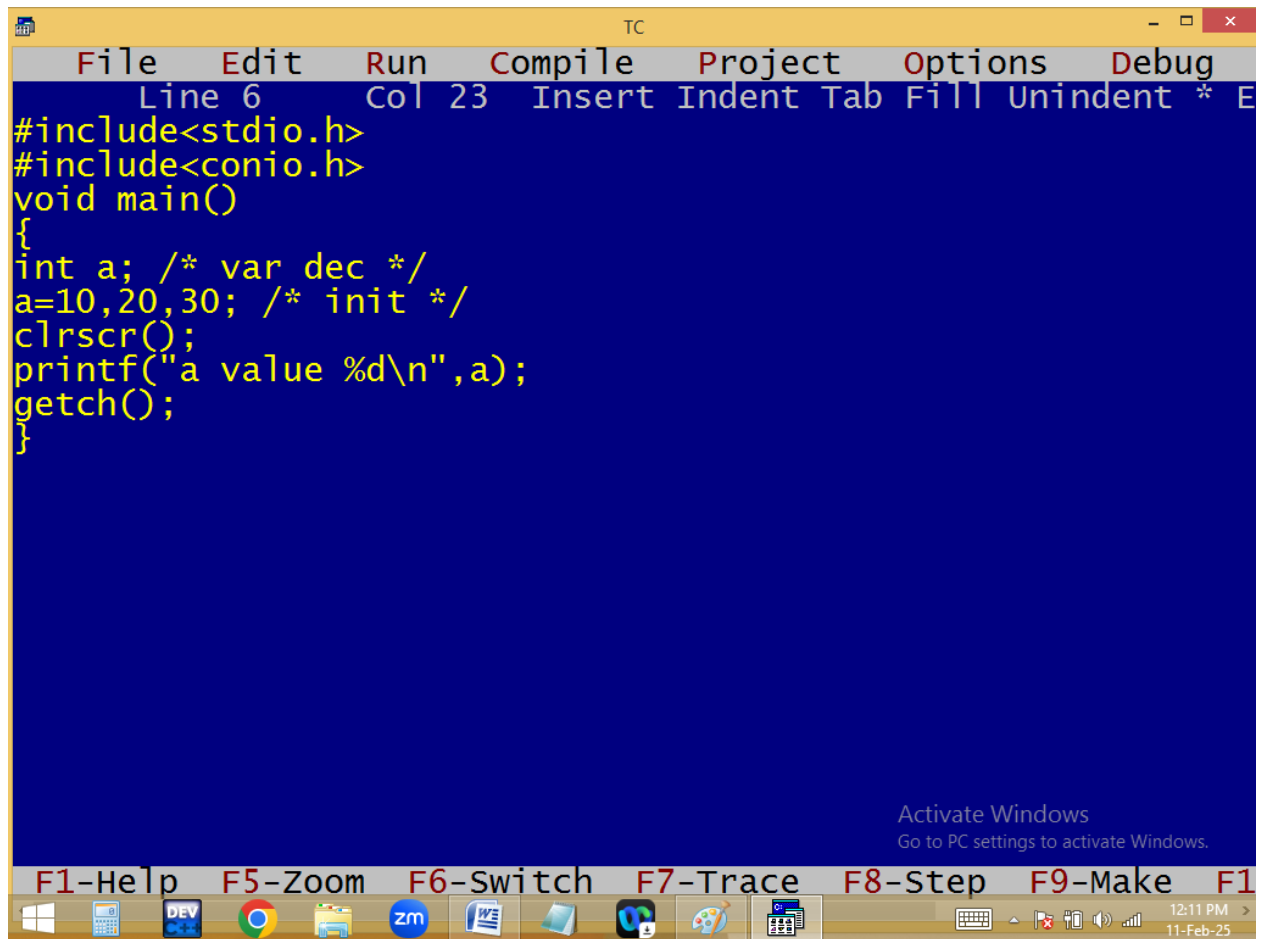
At the bottom of the IDE window, there is a text prompt: "Activate Windows. Go to PC settings to activate Windows." Below the IDE window is the Windows taskbar. It features a row of function key shortcuts: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F10-Exit. The taskbar also includes icons for the Start menu, Task View, File Explorer, Google Chrome, Zoom, Word, and several other applications. The system tray on the right shows the time as 12:06 PM and the date as 11-Feb-25.



int a = (10, 20 , 30);



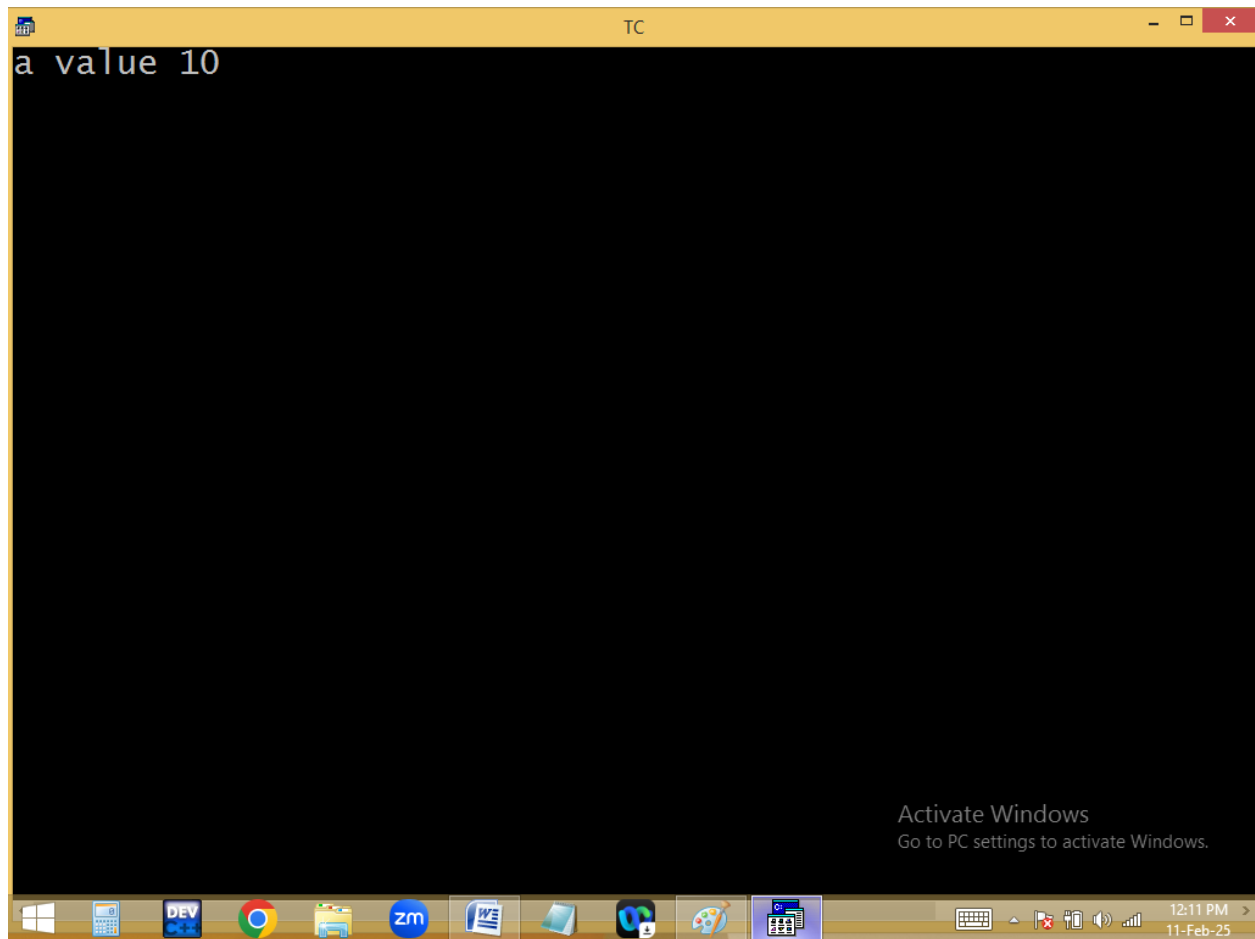
~~a=10~~ ✓
~~a=20~~
a=30
p(a)



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 6 Col 23" and provides editing options: Insert, Indent, Tab, Fill, Unindent, and *. The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a; /* var dec */
a=10,20,30; /* init */
clrscr();
printf("a value %d\n",a);
getch();
}
```

An "Activate Windows" watermark is visible in the bottom right of the IDE window, stating "Go to PC settings to activate Windows." Below the IDE window is the Windows taskbar. It features a row of function key shortcuts: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F10-Run. The taskbar includes icons for the Start menu, Task View, File Explorer, Google Chrome, Zoom, Word, and several other applications. The system tray on the right shows the time as 12:11 PM and the date as 11-Feb-25.




a = 10, ~~20~~ 30;

```
TC
File Edit Run Compile Project Options Debug
Line 11 Col 11 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a; /* var dec */
a=10,(20),30; /* init */
clrscr();
printf("a value %d\n",a);
getch();
}
/* a=10 */_

Activate Windows
Go to PC settings to activate Windows.

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Exit
```

a = 10 , (20) , 30;

a = 10 , 20 , 30 ;

```
TC
File Edit Run Compile Project Options Debug
Line 6 Col 13 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a; /* var dec */
a=10,(20,30); /* init */
clrscr();
printf("a value %d\n",a);
getch();
}
/* a=10 */

Activate Windows
Go to PC settings to activate Windows.

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Exit
```

a = 10, (20, 30);



```
TC
File Edit Run Compile Project Options Debug
Line 11 Col 7 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a; /* var dec */
a=(10,20),(30,40); /* init */
clrscr();
printf("a value %d\n",a);
getch();
}
/* a=20 */

Activate Windows
Go to PC settings to activate Windows.

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Exit
```

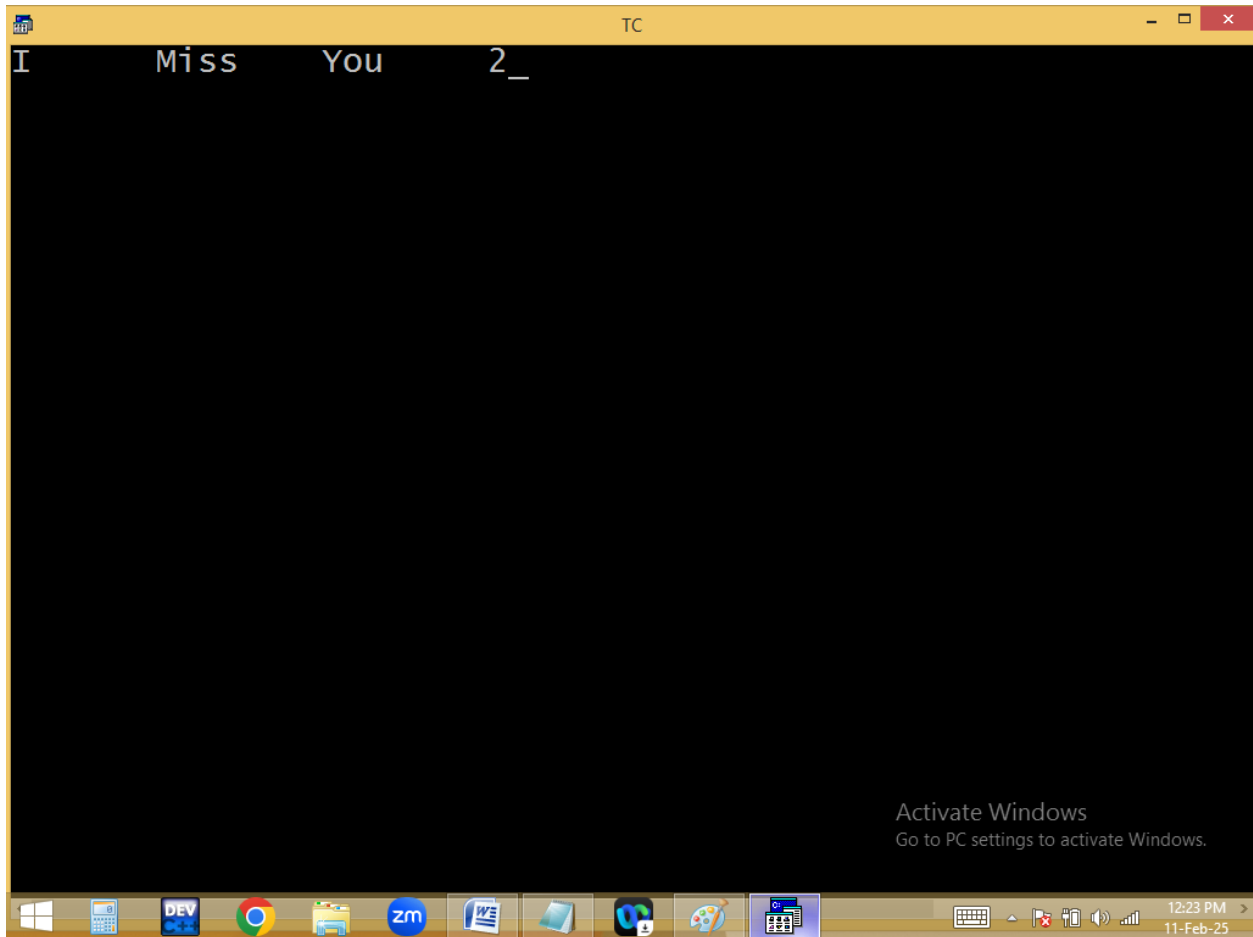
a = (10, 20), (30, 40);

→ →

a = 20, 40

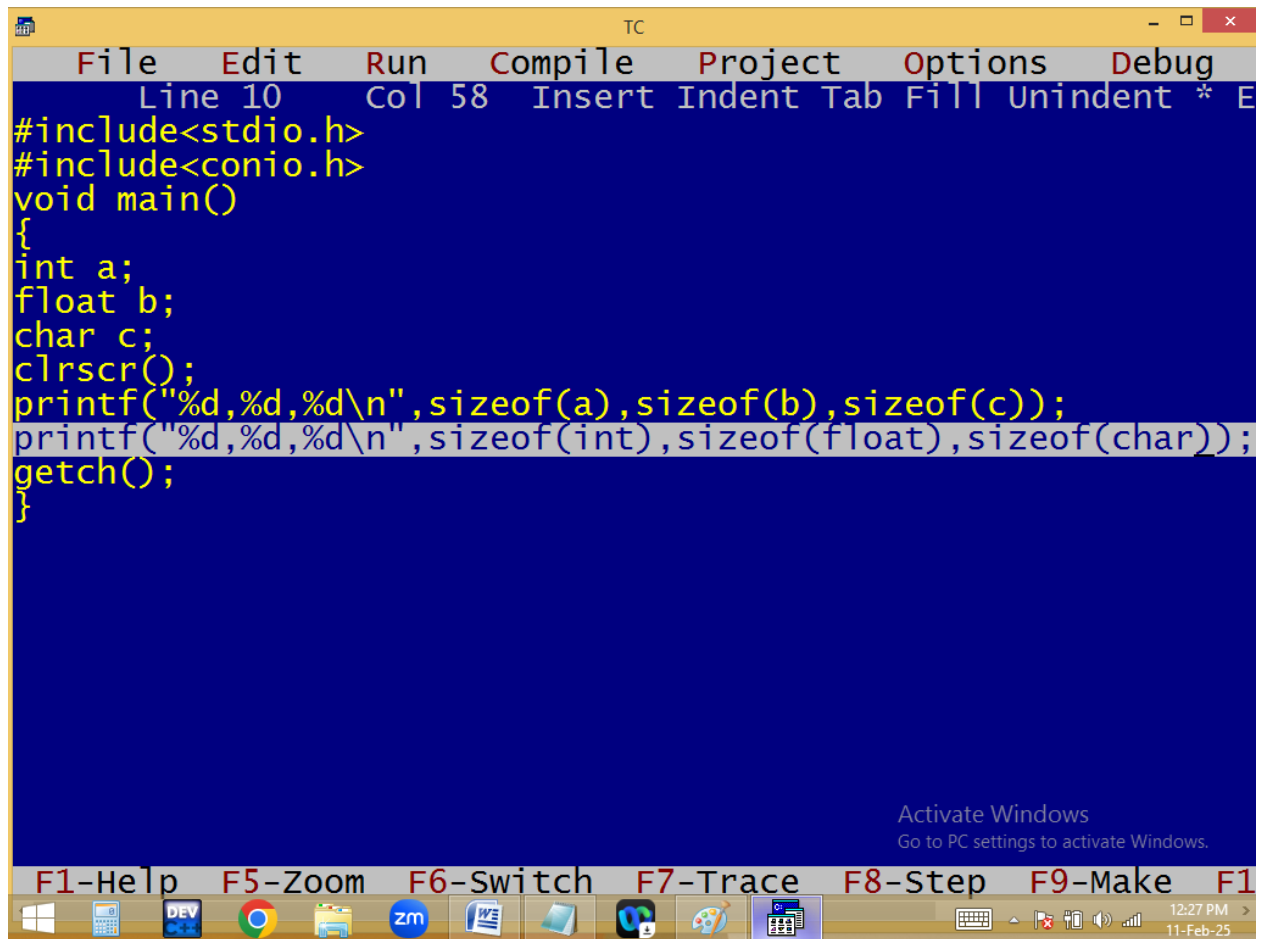
↖

```
TC
File Edit Run Compile Project Options Debug
Line 11 Col 1 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a; /* var dec */
clrscr();
a= printf("I\t"), printf("Miss\t"), printf("You\t");/*init*/
printf("%d",a);
getch();
}
—
Activate Windows
Go to PC settings to activate Windows.
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run
12:22 PM
11-Feb-25
```




a = 2 , 5, 4;

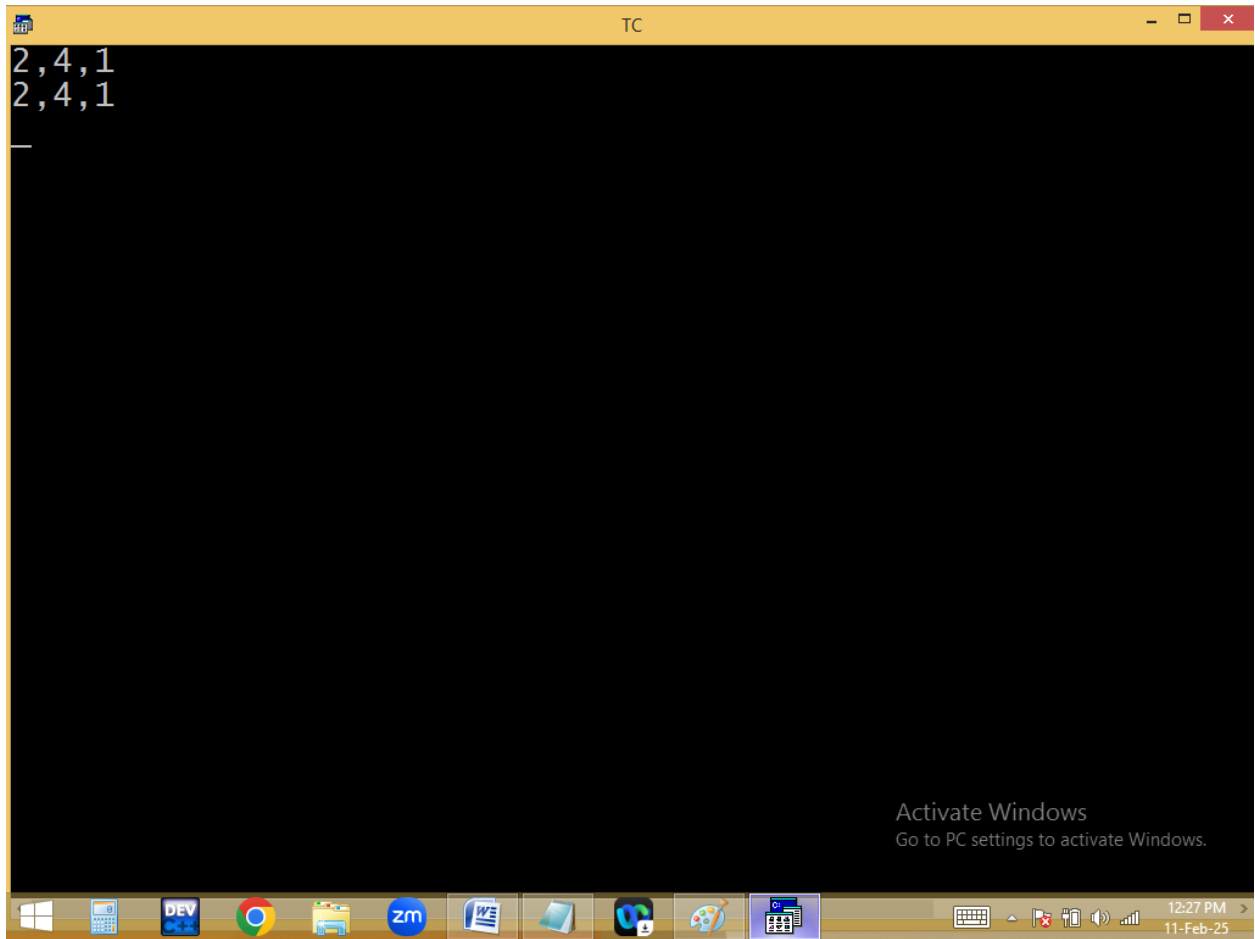
sizeof(): It return the no of bytes taken by a variable / data type / value.



The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status line indicates "Line 10 Col 58" and provides keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", and a cursor icon. The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a;
float b;
char c;
clrscr();
printf("%d,%d,%d\n",sizeof(a),sizeof(b),sizeof(c));
printf("%d,%d,%d\n",sizeof(int),sizeof(float),sizeof(char));
getch();
}
```

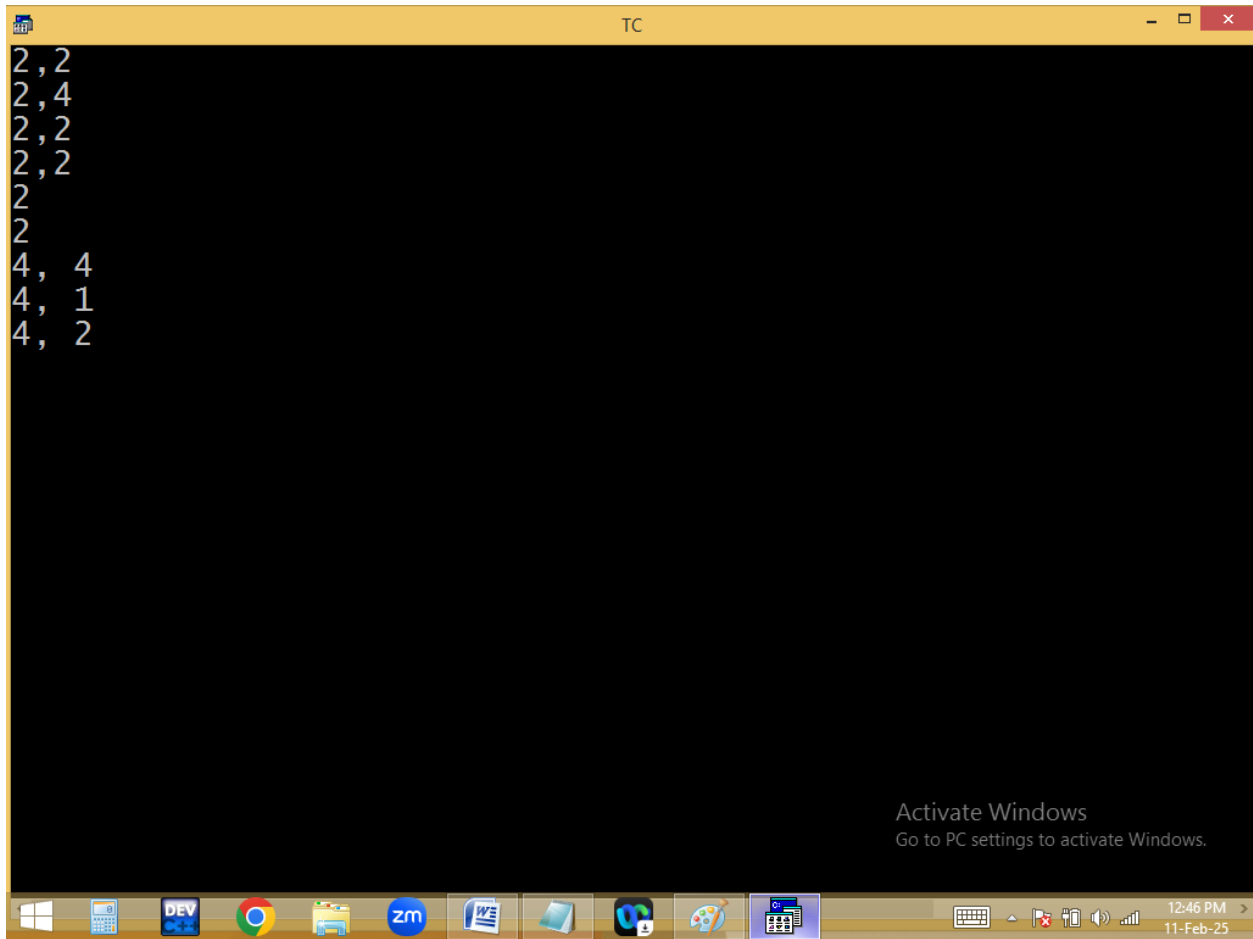
At the bottom of the IDE window, there is a footer with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". Below the IDE window is the Windows taskbar, which includes icons for the Start menu, Task View, File Explorer, Google Chrome, Zoom, Word, and several other applications. The system tray on the right shows the time as "12:27 PM" and the date as "11-Feb-25". An "Activate Windows" watermark is visible in the bottom right corner of the desktop area.

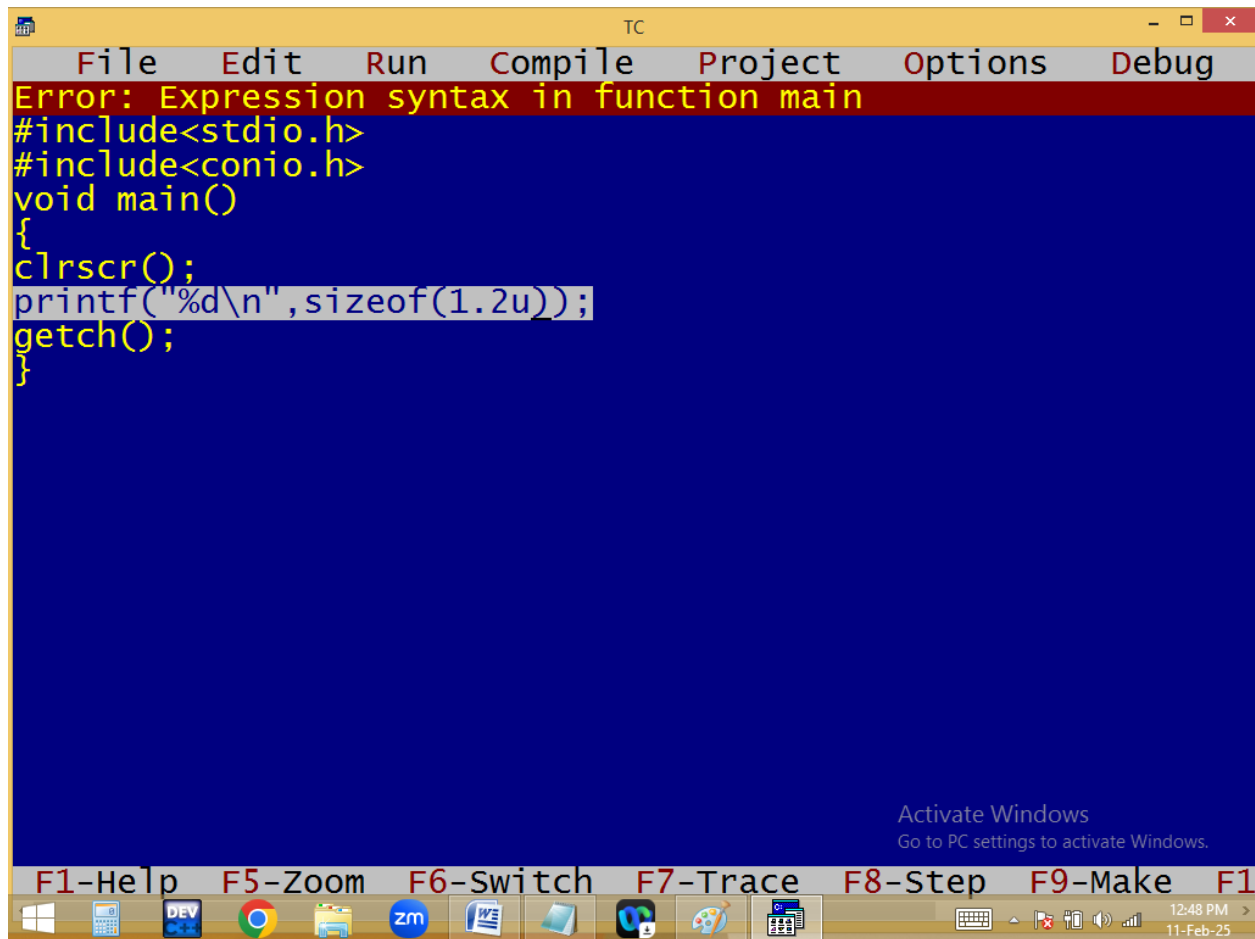



```
TC
File Edit Run Compile Project Options Debug
Line 17 Col 1 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
unsigned int a=32768; int b=32768;
clrscr();
printf("%d,%d\n",sizeof(100),sizeof(-100));
printf("%d,%d\n",sizeof(32767),sizeof(32768));
printf("%d,%d\n",sizeof(a),sizeof(b));
printf("%d,%d\n",sizeof(32768u),sizeof(32768U));/*suffix*/
printf("%d\n",sizeof((int)32768));/*prefix type casting*/
printf("%d\n",sizeof((unsigned)32768));
printf("%d, %d\n",sizeof(100L), sizeof(100l));
printf("%d, %d\n",sizeof((float)100), sizeof((char)100));
printf("%d, %d\n",sizeof(70000u), sizeof((unsigned)70000));
getch();
}

/* after 32767 every int is long int i.e. 4 bytes */

Activate Windows
Go to PC settings to activate Windows.
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run
```

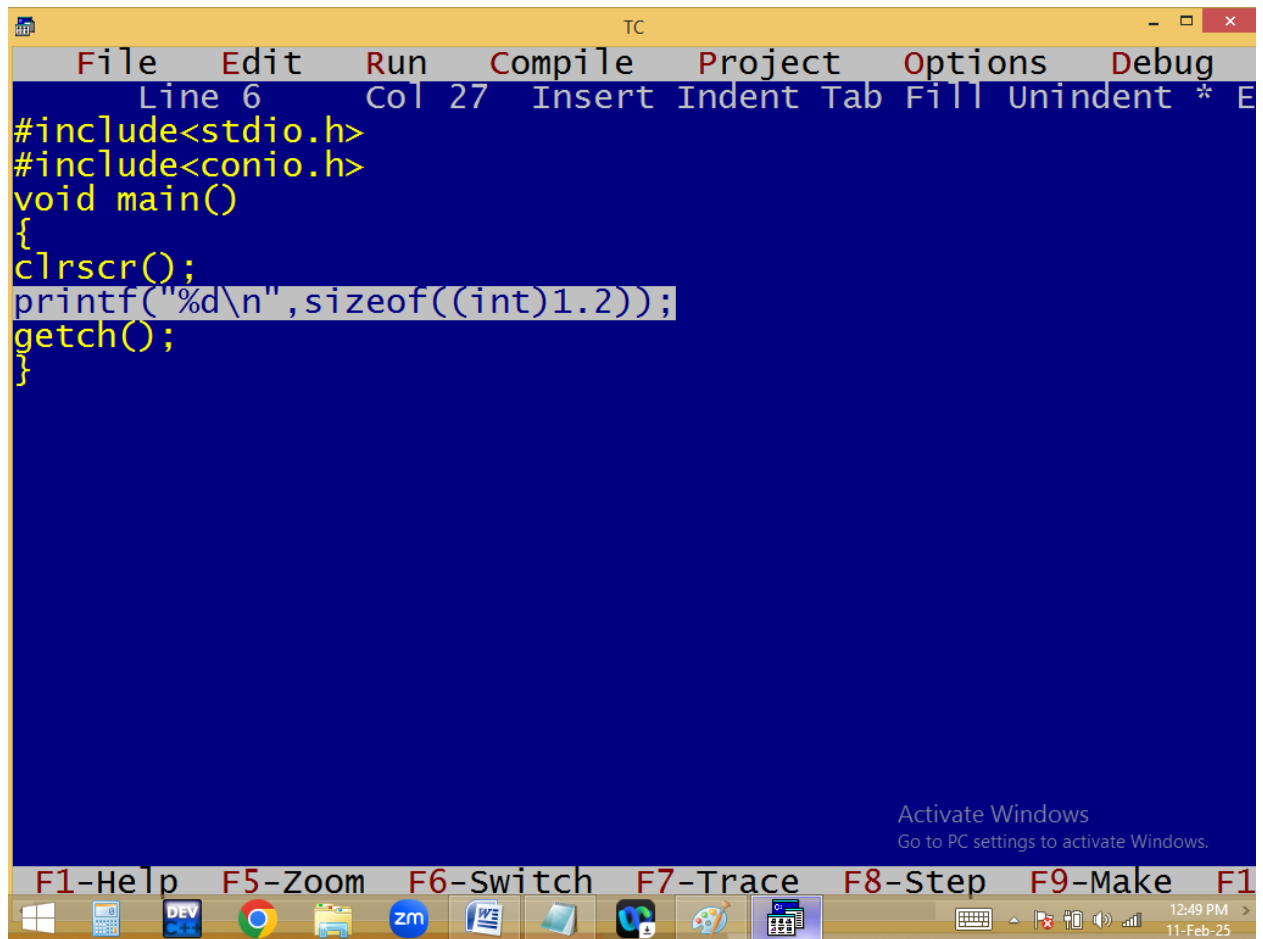




The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error message banner at the top reads "Error: Expression syntax in function main". The code editor has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",sizeof(1.2u));
getch();
}
```

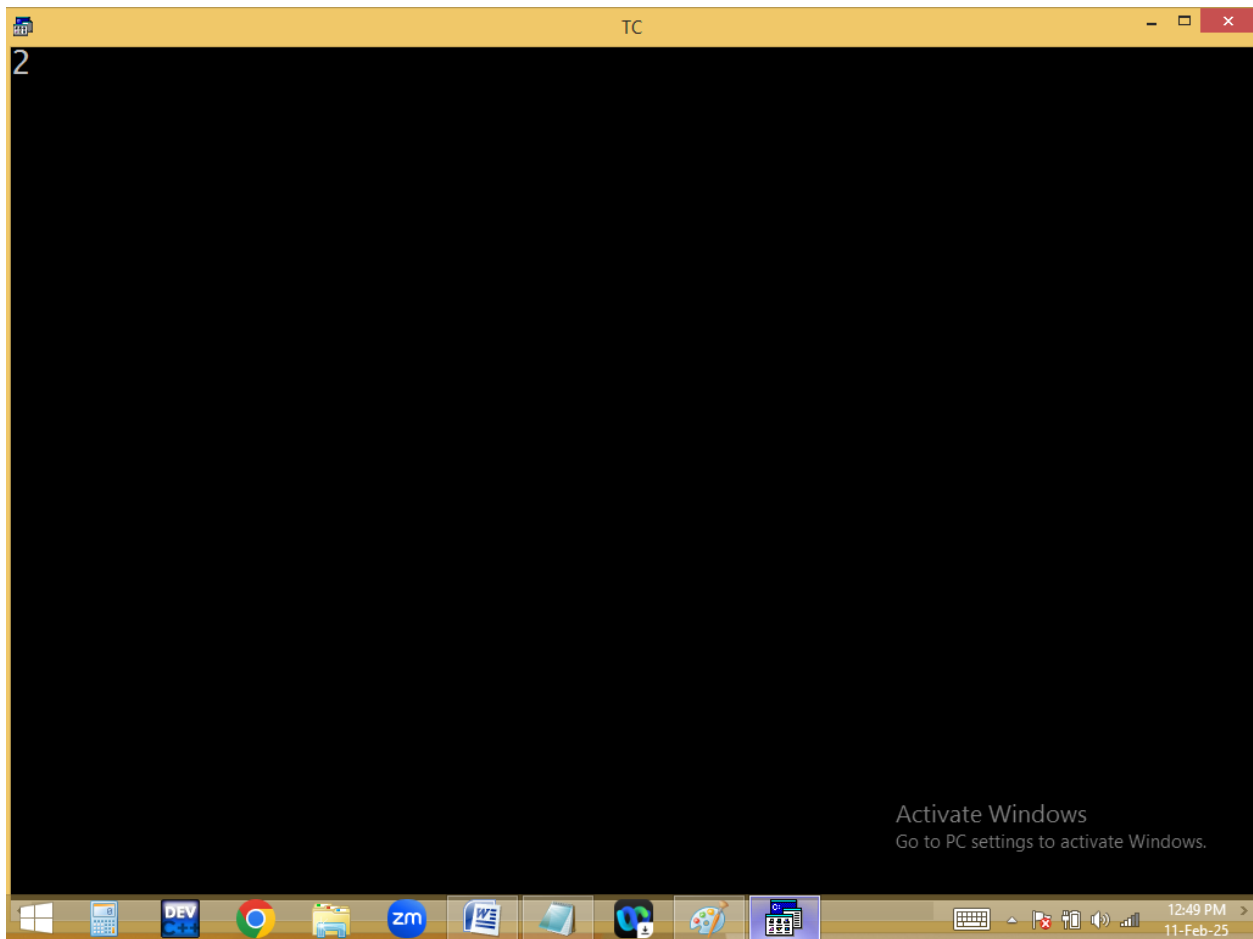
The line `printf("%d\n",sizeof(1.2u));` is highlighted with a light blue selection. At the bottom of the IDE window, there is a status bar with the text "Activate Windows Go to PC settings to activate Windows." and a row of function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar is visible at the very bottom, showing icons for the Start menu, Task View, File Explorer, and several other applications. The system clock in the bottom right corner shows "12:48 PM" and "11-Feb-25".

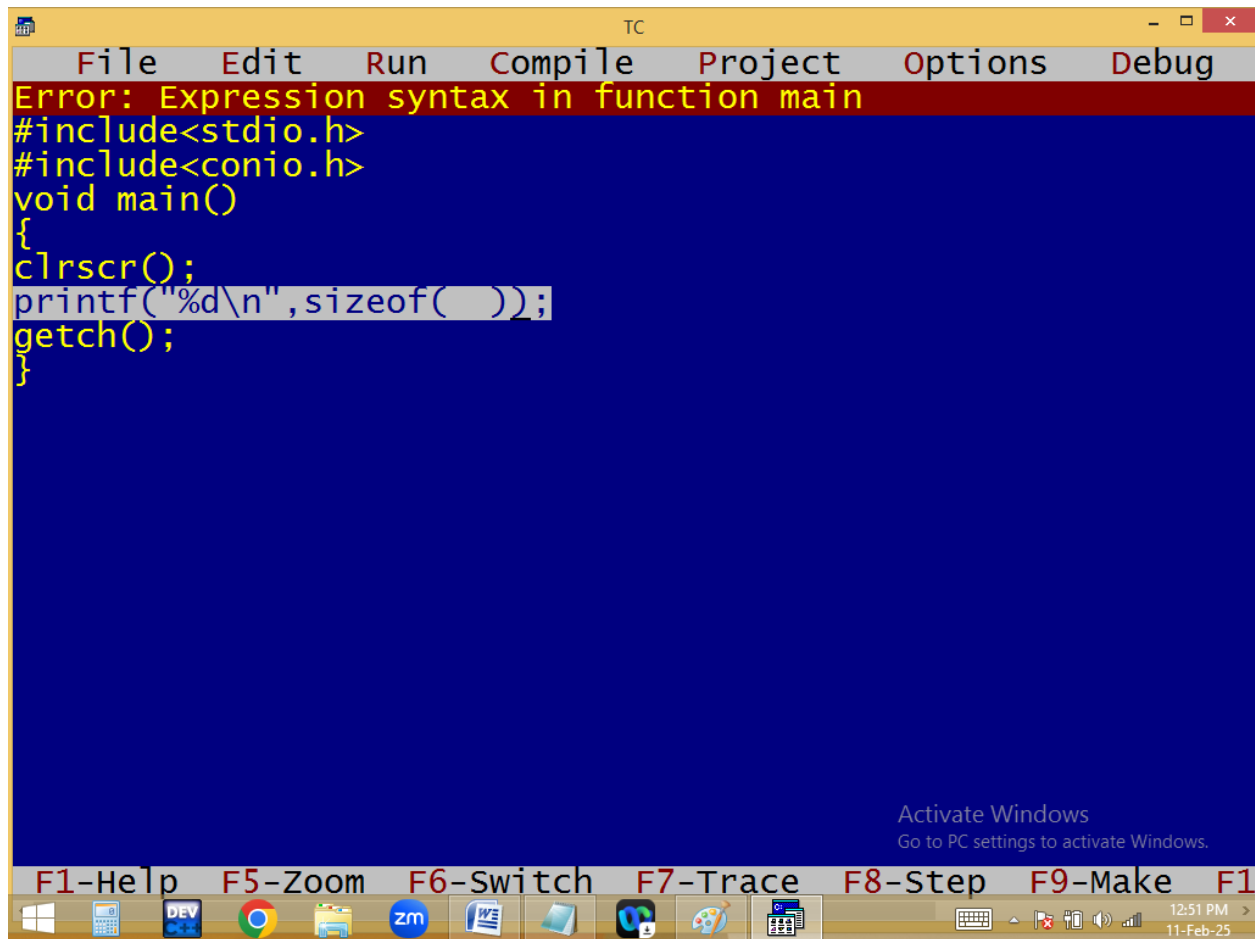


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 6", "Col 27", and lists editing actions: Insert, Indent, Tab, Fill, Unindent, and *. The code editor has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",sizeof((int)1.2));
getch();
}
```

The line `printf("%d\n",sizeof((int)1.2));` is highlighted with a light blue background. At the bottom of the code editor, there is a message: "Activate Windows Go to PC settings to activate Windows." Below the code editor is a toolbar with function key shortcuts: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F10-Run. The Windows taskbar is visible at the very bottom, showing icons for the Start menu, Task View, File Explorer, Microsoft Edge, and several other applications. The system clock in the bottom right corner shows "12:49 PM" and "11-Feb-25".





The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error message banner at the top reads "Error: Expression syntax in function main". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",sizeof( ));
getch();
}
```

The line `printf("%d\n",sizeof( ));` is highlighted, indicating the source of the syntax error. The error occurs because the `sizeof` operator is followed by an opening parenthesis without a closing one. The IDE also features a status bar at the bottom with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". An "Activate Windows" watermark is visible in the bottom right corner of the IDE window. The Windows taskbar at the very bottom shows the Start button, taskbar icons for various applications, and the system clock displaying "12:51 PM" and "11-Feb-25".

